



Integrating motion analysis into smart coaching to improve physical literacy and movement quality in novice athletes

Rahma Dewi*

Universitas Negeri Medan,
INDONESIA

Ardi Nusri

Universitas Negeri Medan,
INDONESIA

Eva Faridah

Universitas Negeri Medan,
INDONESIA

Bessy Sitorus Pane

Universitas Negeri Medan,
INDONESIA

Muhammad Reza Destya

Universitas Negeri Medan,
INDONESIA

Article Info

Article history:

Received: March 4, 2026

Revised: May 27, 2026

Accepted: September 10, 2026

Keywords:

Augmented feedback;
Markerless motion analysis;
Movement quality;
Physical literacy;
Smart coaching.

Abstract

Background: Markerless motion analysis can enrich coaching feedback with objective movement information, but evidence linking it to both physical literacy and functional movement outcomes in novice athletes remains limited.

Aims: This study examined whether an eight-week Smart Coaching program supported by markerless motion analysis was associated with greater short-term change than conventional coaching.

Methods: A nonrandomized quasi-experimental pretest-posttest control-group design involved 60 novice athletes aged 12-15 years (30 per group). Both groups trained three times weekly for eight weeks. Physical literacy was assessed with an adapted CAPL-2 composite and movement quality with the Functional Movement Screen (FMS). Separate 2×2 mixed ANOVAs tested Time $\times$ Group effects.

Results: Smart Coaching increased CAPL-2 scores from 55.03 ± 6.69 to 73.64 ± 6.51 and FMS scores from 11.37 ± 1.92 to 16.60 ± 1.54 . Corresponding gains were 18.61 and 5.23 points, compared with 7.64 and 1.43 points under Conventional Coaching. Time $\times$ Group interactions were significant for physical literacy, $F(1, 58) = 48.31$, $p < .001$, partial $\eta^2 = .454$, and FMS, $F(1, 58) = 77.20$, $p < .001$, partial $\eta^2 = .571$. Between-group differences in change were 10.97 points (95% CI 7.81-14.13) and 3.80 points (95% CI 2.93-4.67), respectively.

Conclusion: In this single-site nonrandomized study, Smart Coaching with markerless motion analysis was associated with larger short-term improvements in both outcomes. Randomized and longitudinal replication is warranted.

To cite this article: Dewi, R., Nusri, A., Faridah, E., Pane, B. S., & Destya, M. R. (2026). Integrating motion analysis into smart coaching to improve physical literacy and movement quality in novice athletes. *Journal of Coaching and Sports Science*, 5(3), 106–120. <https://doi.org/10.58524/jcss.v5i3.1468>

This article is licensed under a [Creative Commons Attribution-ShareAlike 4.0 International License](https://creativecommons.org/licenses/by-sa/4.0/) ©2026 by author/s

INTRODUCTION

Digital technologies are increasingly shifting sport coaching from purely observational judgment toward data-informed practice, where movement can be captured, quantified, and revisited during the learning process. Contemporary sports biomechanics has increasingly moved toward more field-deployable measurement systems, while markerless computer-vision approaches extend motion analysis beyond conventional laboratory environments (Yeadon & Pain, 2023; Adlou et al., 2025; Noorbhai et al., 2025). This transition is particularly relevant in developmental coaching, where novice athletes often require repeated, interpretable feedback on fundamental movement patterns rather than highly specialized performance analytics.

Markerless motion capture is promising but should not be treated as a uniform technology. Validation studies show that performance depends on the system, task, joint, plane of movement, camera configuration, and processing pipeline. For example, smartphone- and deep-learning-based systems have produced useful estimates in selected squatting, jumping, gait, and return-to-sport tasks, whereas errors may increase for particular planes and highly dynamic field movements; competition-based testing has also demonstrated that pose-estimation performance can degrade

* Corresponding author:

Dewi, R., Universitas Negeri Medan, INDONESIA. ✉rahmadewi@unimed.ac.id.

outside controlled settings (Aderinola et al., 2023; Turner et al., 2024; Bae et al., 2024; Huang et al., 2024; Lima et al., 2024; Cronin et al., 2024). Recent systematic reviews therefore recommend system- and task-specific validation rather than assuming that markerless outputs are interchangeable with laboratory motion capture (Yoma et al., 2025; Ogura et al., 2026; Varcin & Boocock, 2026).

The pedagogical value of motion analysis depends not only on measurement accuracy but also on how the information is converted into feedback. Augmented visual and quantitative feedback can direct attention toward movement features that may be difficult to perceive intrinsically. Systematic reviews and meta-analyses indicate that feedback can enhance motor-skill learning and training performance, although effects vary with feedback modality, frequency, task complexity, and population; evidence from children with coordination difficulties likewise cautions that technology-supported feedback is not uniformly superior to conventional delivery (Möding et al., 2022; Tissera et al., 2022; van der Veer et al., 2022; Weakley et al., 2023; Welsby et al., 2024). These findings support a coaching model in which coaches interpret motion data as learning cues rather than treating them as an autonomous decision-maker.

Physical literacy provides a broader developmental lens for evaluating such an intervention. Physical literacy is multidimensional: it incorporates physical competence together with motivational/affective and cognitive resources that support engagement in physical activity. Intervention evidence suggests that physical competence often shows the largest gains, while motivation, confidence, knowledge, and activity behavior can also change when programs deliberately address those dimensions (Carl et al., 2022; Barnett et al., 2023; Jerebine et al., 2024). Motor competence in adolescence is also associated with physical activity, fitness, and psychosocial characteristics, underscoring why movement-focused coaching may have developmental relevance beyond a single technical skill (Burton et al., 2023; Coe et al., 2024).

At the same time, measurement must match the age and construct being studied. CAPL-2 was originally developed for children aged 8-12 years, and assessment reviews emphasize the importance of population-specific validity when extending physical-literacy tools to older adolescents (Longmuir et al., 2018; Barnett et al., 2023). Adolescent-specific measures such as the PPLA-O/PPLA-Q illustrate that physical-literacy assessment in older youth requires explicit attention to age, cognitive content, and context (Mota et al., 2022, 2023). Accordingly, the present study treats the CAPL-2-derived composite as a continuous study outcome and does not apply age-normative categorical interpretations beyond the evidence available for the original instrument.

We evaluated movement quality separately using the Functional Movement Screen (FMS). FMS provides a structured seven-task assessment of fundamental movement patterns, and intervention reviews indicate that total FMS scores can improve following targeted movement programs; broader physical-activity interventions also support gains in motor competence among children, although outcomes vary by intervention design and population (Clark et al., 2022; Huang et al., 2022; Moon et al., 2024). FMS total scores should not be interpreted as a direct biomechanical diagnosis or a stand-alone measure of injury risk. In adolescent populations, the relationship between FMS and athletic performance is heterogeneous, further supporting the use of FMS here as an operational measure of functional movement quality rather than a proxy for overall athletic performance (Fitton Davies et al., 2022; O'Brien et al., 2022).

The specific research gap lies at the intersection of these literatures. Existing markerless-motion research is dominated by validation and biomechanical measurement questions, while physical-literacy interventions are typically evaluated without a technology-supported motion-feedback loop. Long-term athlete-development literature also emphasizes that youth development is multidimensional and that short-term physical gains cannot, by themselves, establish sustained developmental benefit (Söker et al., 2025). Consequently, limited evidence addresses whether embedding markerless motion cues within routine coaching is associated with simultaneous change in a holistic physical-literacy composite and functional movement quality among novice, multisport athletes (Philipp et al., 2024; Yoma et al., 2025; Adlou et al., 2025). The novelty of this study is therefore not the use of motion capture alone, but integrating objective movement cues into a Smart Coaching feedback process while evaluating two developmentally distinct outcomes in a novice-athlete context.

Conceptually, Smart Coaching followed a feedback pathway in which movement was recorded, an AI-assisted markerless system generated kinematic cues, the coach interpreted those cues in

relation to the movement task, and the athlete received visual or quantitative information followed by corrective practice. The authors did not document the specific pose-estimation architecture; accordingly, the technological component is treated as an assistive measurement layer rather than an autonomous decision system. This pathway is expected to influence FMS most directly through movement-error recognition and correction. Potential effects on physical literacy are broader and more indirect: improved movement competence can contribute to perceived capability, while visual explanation may support knowledge and understanding; motivation, confidence, and activity behavior require additional pedagogical mechanisms and should not be assumed to change solely because motion analysis is present (Carl et al., 2022; Weakley et al., 2023; Jerebine et al., 2024).

Accordingly, this study examined two prespecified hypotheses. H1: The Smart Coaching group would demonstrate a greater pretest-to-posttest increase in the CAPL-2-based physical-literacy composite than the Conventional Coaching group. H2: The Smart Coaching group would demonstrate a greater pretest-to-posttest increase in FMS total score than the Conventional Coaching group. Claims regarding motor-learning rate, long-term athlete development, injury prevention, or sustained physical-activity behavior were treated as theoretical implications rather than measured outcomes.

METHOD

Research Design and Setting

This study employed a quantitative, nonrandomized quasi-experimental pretest-posttest control-group design. The study involved two training conditions: Smart Coaching supported by markerless motion analysis ($n = 30$) and Conventional Coaching ($n = 30$). The intervention lasted eight weeks, with three approximately 90-minute sessions per week (24 scheduled sessions; approximately 36 hours of planned training exposure). Random allocation was not documented; accordingly, we treat the design analytically as nonrandomized. The Group $\times$ Time interaction is therefore interpreted as evidence of differential change under the observed training conditions rather than as proof of a randomized causal effect.

Table 1. Study Design and Operational Contrast between Coaching Conditions

Component	Smart Coaching + Motion Analysis	Conventional Coaching
Duration and schedule	8 weeks; 3 sessions/week; ~90 min/session	8 weeks; 3 sessions/week; ~90 min/session
Movement assessment during coaching	Markerless video analysis used to derive kinematic cues for coach interpretation; the specific pose-estimation architecture was not documented	Coach observation without markerless-analysis output
Feedback source	Coach interpretation supported by visual/quantitative movement cues	Coach observation and verbal feedback
Correction process	Movement deviations identified from observed and system-derived cues, followed by corrective practice	Movement corrections based on coach observation and verbal instruction
Primary outcomes	CAPL-2-based composite and FMS total score	CAPL-2-based composite and FMS total score
Exposure control	Same planned session frequency and duration	Same planned session frequency and duration

Table 1 clarifies that motion analysis functioned as an additional information layer within coaching rather than as a separate outcome. The exact capture frequency within each session, progression of individual corrective exercises, and administrative allocation mechanism were not documented. We acknowledge these implementation constraints when interpreting reproducibility and internal validity.

Participants and Allocation

The analytic sample comprised 60 novice athletes aged 12-15 years, purposively recruited from a regional training center in Medan City, Indonesia. Athletes represented athletics, badminton, taekwondo, and soccer. Inclusion criteria included registration as a novice athlete at the training

center, less than two years of formal competitive training, no musculoskeletal injury in the preceding six months, regular participation in scheduled training, and written agreement to participate from the athlete, along with written informed consent from a parent or legal guardian. We analyzed 30 athletes in each coaching condition. The report did not document random allocation or matching, and group-specific sex distribution, mean age, sport distribution, and maturation status were unavailable; residual baseline imbalance therefore cannot be excluded.

All 60 participants included in the analysis had complete pretest and posttest outcome records. The report did not document the number initially screened or enrolled, reasons for any pre-analytic exclusion, or session-level attendance; therefore, it describes an analytic sample of 60 without inferring zero attrition. The authors also did not document an a priori sample-size calculation, eligible-population size, or recruitment denominator.

Research Ethics

The authors' institutional research ethics committee approved the study protocol (approval no. 025/OR/PL.1/07/2025) before participant recruitment and data collection. Because participants were minors, each athlete provided written agreement to participate, and a parent or legal guardian provided written informed consent. No participant-identifying images or video frames are presented in this manuscript.

Instrumentation and Outcome Measures

The study distinguished primary outcomes from intervention-support variables. Physical literacy was represented by the CAPL-2-based composite score (0-100), whereas movement quality was represented by the FMS total score (0-21). Markerless kinematic variables were used solely to support coaching feedback and were not mathematically combined with FMS or analyzed as a third primary outcome.

Table 2. Measurement Roles, Scales, and Interpretive Boundaries

Construct/role	Measure	Scale	Role in analysis	Interpretive boundary
Physical literacy	CAPL-2-based composite	0-100	Primary outcome	Used as a continuous composite; age-specific norms for 12-15 years were not assumed
Movement quality	Functional Movement Screen (FMS)	0-21	Primary outcome	Total score only; no unsupported “excellent/good/moderate/poor” categories
Motion-analysis feedback	AI-assisted markerless kinematic cues	Continuous units	Intervention-support variable	Used to inform feedback; not combined with FMS and not reported as a primary endpoint

Table 2 makes explicit that inferential conclusions concern only the two primary outcomes. Markerless data served as diagnostic and feedback cues and were not analyzed as independent longitudinal endpoints.

Physical Literacy Assessment

Physical literacy was assessed using a CAPL-2-based composite structure covering physical competence, daily physical-activity behavior, motivation and confidence, and knowledge and understanding. CAPL-2 was originally validated for children aged 8-12 years (Longmuir et al., 2018). Because this study included athletes up to age 15, the composite is reported as an adapted study measure rather than as a fully validated age-normed CAPL-2 classification. Contemporary reviews identify age- and population-specific validity as a central issue in physical-literacy assessment and document adolescent-specific alternatives (Barnett et al., 2023; Mota et al., 2022, 2023). We make no claim of de novo construct validation in the present sample because item-level validation data were not available.

Table 3. CAPL-2-Based Domains Represented in the Physical-Literacy Composite

Domain	Indicators/components represented	Score contribution
Physical competence	PACER, plank, CAMSA, anthropometry	0-30
Daily physical-activity behavior	Step counts, MVPA, self-report/activity record	0-30
Motivation and confidence	Intrinsic motivation, confidence, enjoyment items	Component of composite
Knowledge and understanding	Physical-activity knowledge items	Component of composite
Total physical literacy	Composite of represented domains	0-100

Only the total physical-literacy composite was available at pretest and posttest; we did not analyze domain-level post-intervention scores. Accordingly, we cannot attribute the overall change to any single physical-literacy domain.

Movement Quality Assessment

We operationalized movement quality as the FMS total score. The FMS comprises seven movement tasks scored from 0 to 3, producing a maximum total score of 21. The analysis treats FMS as a continuous total score. It does not apply categorical labels such as excellent, good, moderate, or poor because such cut-offs were not prespecified or supported by a validated source for this sample. Reviews of functional movement interventions show that total FMS can respond to targeted programs, but interpretation should remain distinct from diagnostic or injury-prediction claims (Clark et al., 2022; Huang et al., 2022).

Table 4. Functional Movement Screen Tasks Used to Operationalize Movement Quality

FMS Task	Primary movement feature	Task score
Deep squat	Bilateral mobility and stability	0-3
Hurdle step	Single-leg stability	0-3
Inline lunge	Dynamic stability	0-3
Shoulder mobility	Upper-body mobility	0-3
Active straight-leg raise	Hamstring and hip mobility	0-3
Trunk-stability push-up	Core/trunk stability	0-3
Rotary stability	Neuromuscular coordination	0-3
Total	Seven-task sum	0-21

Table 4 defines the FMS outcome used throughout the inferential analysis. Biomechanical variables from markerless motion analysis were not added to this score.

Procedures and Motion-Analysis Feedback Pipeline

The intervention used an AI-assisted markerless motion-analysis system with high-speed digital video acquisition at least 120 frames/s. Operationally, the workflow included image acquisition, markerless pose estimation, extraction of movement descriptors, visualization of movement deviations, and provision of quantitative cues for coaches to interpret. Movement descriptors included joint angles, range of motion, step length, ground-contact time, center-of-mass displacement, movement velocity, and symmetry. We did not include a composite Technical Efficiency Index because the authors did not document a prespecified formula, validation basis, weighting scheme, or analytical role.

The software name/version, pose-estimation architecture, camera number and geometry, calibration procedure, and system-specific validation study were not documented. Because markerless validity is system- and task-dependent, we assign no generic validity coefficient to the present system. We therefore use contemporary validation literature only to contextualize field utility and known limitations across markerless approaches (Turner et al., 2024; Lima et al., 2024; Bae et al., 2024; Philipp et al., 2024; Yoma et al., 2025).

Table 5. Markerless Motion Descriptors Used as Coaching-Feedback Cues

Descriptor	Unit	Coaching Interpretation
Hip joint angle	degrees	Hip flexion/extension pattern
Knee joint angle	degrees	Knee alignment and excursion

Descriptor	Unit	Coaching Interpretation
Ankle joint angle	degrees	Dorsiflexion/plantarflexion pattern
Range of motion	degrees	Joint excursion
Step length	cm	Spatial step characteristic
Ground contact time	ms	Temporal foot-ground interaction
Center-of-mass displacement	cm	Body-motion/stability cue
Movement velocity	m/s	Movement-speed cue
Movement symmetry	%	Right-left symmetry cue

These descriptors supported the coaching feedback loop but were not independent endpoints. Their role is therefore limited to intervention support; outcome changes are not attributed directly to biomechanical variables that were not analyzed longitudinally.

Analysis Plan

Analyses were conducted in IBM SPSS Statistics 29.0. Descriptive statistics summarized means and standard deviations for each primary outcome at pretest and posttest. Shapiro-Wilk tests yielded $p = .056-.698$ across analyzed datasets, and Levene tests were non-significant ($p > .05$). These diagnostics supported retention of the planned parametric model. Because Time had only two levels, sphericity was not applicable.

A 2×2 mixed-design ANOVA was used separately for physical literacy and FMS, with Time (pretest, posttest) as the within-participant factor and Group (Smart Coaching, Conventional Coaching) as the between-participant factor. The primary inferential quantity was the Time $\times$ Group interaction. Effect size for ANOVA effects was partial eta squared (partial η^2), reported consistently throughout. The interaction degrees of freedom were 1 and 58. Given the nonrandomized design, interpret statistical significance alongside baseline group differences and study limitations; a non-significant baseline test does not prove equivalence.

For descriptive precision, we calculated 95% confidence intervals around each time-specific group mean from the reported SD, $n = 30$, and the t distribution with 29 degrees of freedom. We calculated between-group standardized differences at pretest and posttest as Cohen's d using the pooled between-participant SD, with approximate 95% confidence intervals based on the standard error of d . Within-group standardized-change effects were not estimated because the pre-post covariance required for a defensible repeated-measures standardization was unavailable. For the interaction contrasts, we recovered 95% confidence intervals for the between-group difference in change from the reported interaction F statistic ($F = t^2$ for a 1-df contrast) and the corresponding standard error. Statistical significance was set at $\alpha = .05$ (two-sided).

Scope and Limitations of the Methodology

The design estimates differential pretest-to-posttest change between the observed coaching conditions but does not provide the causal control of random allocation. The authors did not document group-allocation details, assessor blinding, intervention-fidelity indicators, participant-level attendance, or full hardware/software specifications. These constraints are incorporated into interpretation.

RESULTS AND DISCUSSION

Results

The analytic sample included 60 athletes with complete pretest and posttest outcome scores, 30 in each coaching condition. Table 6 summarizes the observed means, standard deviations, and unstandardized gains. Both groups increased on both outcomes, but the absolute gains were larger under Smart Coaching.

Table 6. Descriptive Statistics for Physical Literacy and Movement Quality

Outcome	Group	Pretest, $M \pm$ SD	Posttest, $M \pm$ SD	Mean Gain
Physical literacy (CAPL-2)	Smart Coaching	55.03 $\pm$ 6.69	73.64 $\pm$ 6.51	18.61

Outcome	Group	Pretest, M $\pm$ SD	Posttest, M $\pm$ SD	Mean Gain
Physical literacy (CAPL-2)	Conventional Coaching	53.80 $\pm$ 6.96	61.44 $\pm$ 7.13	7.64
Movement quality (FMS)	Smart Coaching	11.37 $\pm$ 1.92	16.60 $\pm$ 1.54	5.23
Movement quality (FMS)	Conventional Coaching	11.97 $\pm$ 1.73	13.40 $\pm$ 1.61	1.43

At pretest, the Smart Coaching group was 1.23 CAPL-2 points higher and 0.60 FMS points lower than the Conventional Coaching group. These differences were small in standardized terms (approximately $d = 0.18$ for CAPL-2 and $d = -0.33$ for FMS) and should be described as baseline differences rather than as confirmed equivalence. By posttest, the between-group differences were 12.20 CAPL-2 points and 3.20 FMS points in favor of Smart Coaching.

Shapiro-Wilk tests yielded $p = .056-.698$, and Levene tests were non-significant ($p > .05$). based on these diagnostics, the planned mixed ANOVAs were retained. Table 7 presents the complete ANOVA effects with consistent partial-eta-squared notation and degrees of freedom.

Table 7. Two-by-Two Mixed ANOVA Results

Outcome	Effect	F(1, 58)	p	Partial η^2
Physical literacy	Time	276.61	< .001	.827
Physical literacy	Group	18.16	< .001	.238
Physical literacy	Time $\times$ Group	48.31	< .001	.454
Movement quality	Time	237.61	< .001	.804
Movement quality	Group	11.44	.001	.165
Movement quality	Time $\times$ Group	77.20	< .001	.571

For H1, the Time $\times$ Group interaction for physical literacy was statistically significant, $F(1, 58) = 48.31$, $p < .001$, partial $\eta^2 = .454$; the between-group difference in change was 10.97 points (95% CI 7.81-14.13). For H2, the Time $\times$ Group interaction for FMS was also statistically significant, $F(1, 58) = 77.20$, $p < .001$, partial $\eta^2 = .571$; the between-group difference in change was 3.80 points (95% CI 2.93-4.67). Thus, the observed pretest-to-posttest trajectories differed between groups for both outcomes, with larger gains under Smart Coaching.

Figure 1 visualizes the interaction pattern for physical literacy using mean scores and 95% confidence intervals calculated from the standard deviations shown in Table 6 and the group size. The figure complements the tabulated results by displaying both change magnitude and uncertainty around each time-specific mean.

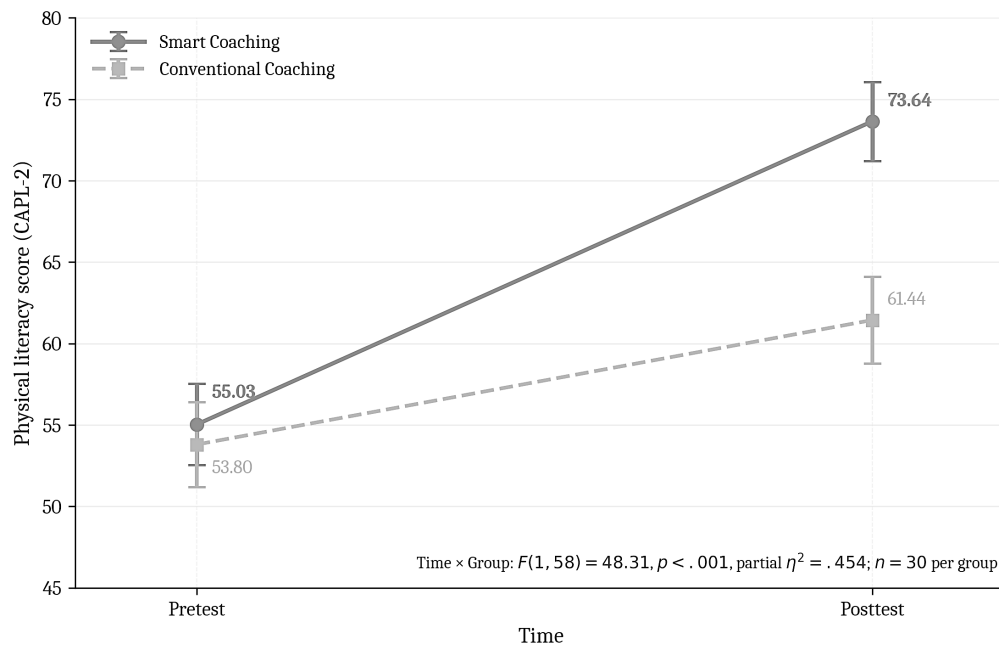


Figure 1. Pretest-to-posttest change in physical literacy (CAPL-2) by coaching condition with 95% confidence intervals. The y-axis is truncated to improve visual resolution.

As shown in Figure 1, the groups were close at baseline but separated markedly at posttest. The steeper increase in the Smart Coaching group is consistent with the significant Time × Group interaction for the CAPL-2-based composite, $F(1, 58) = 48.31, p < .001, \text{partial } \eta^2 = .454$.

Figure 2 presents the corresponding interaction pattern for movement quality. Plotting the FMS means with 95% confidence intervals makes the differential trajectory more interpretable than a simple pretest-posttest bar chart.

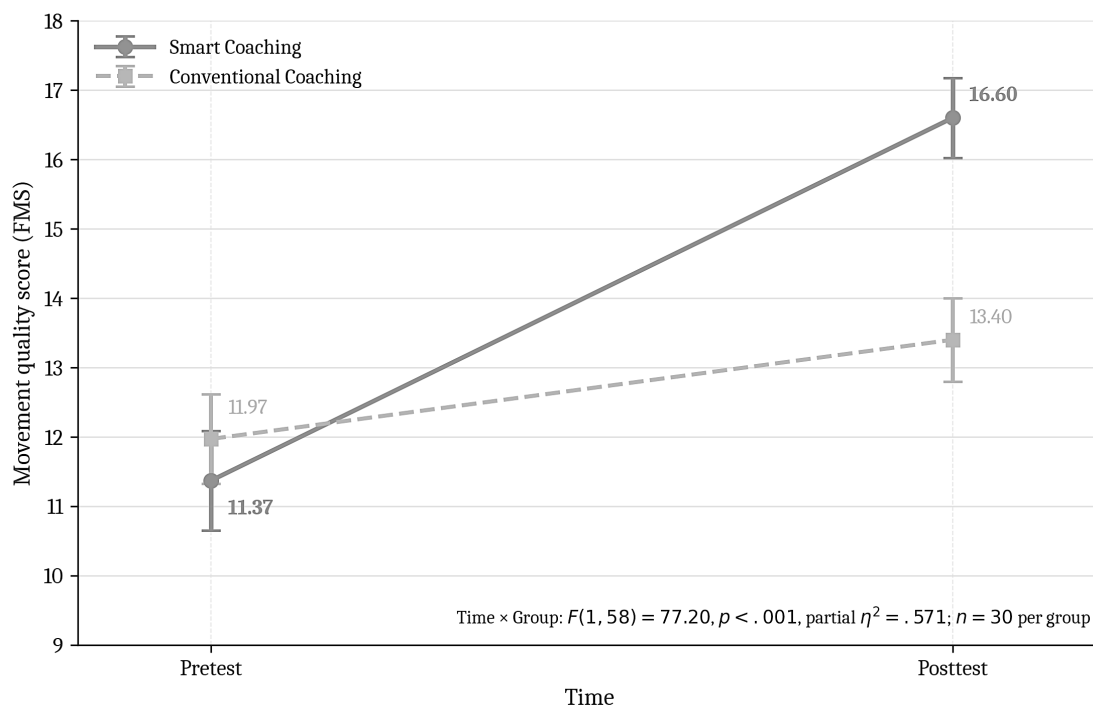
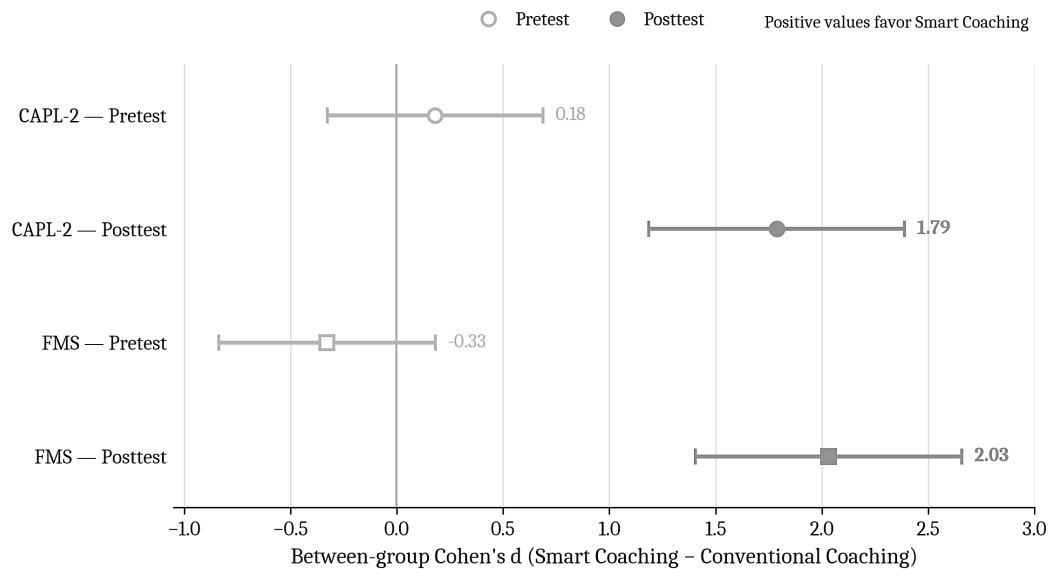


Figure 2. Pretest-to-posttest change in Functional Movement Screen scores by coaching condition with 95% confidence intervals. The y-axis is truncated to improve visual resolution.

Figure 2 shows a substantially larger increase in the Smart Coaching group, whereas the Conventional Coaching trajectory is comparatively modest. This descriptive pattern aligns with the significant Time $\times$ Group interaction for FMS, $F(1, 58) = 77.20$, $p < .001$, partial $\eta^2 = .571$.

To contextualize between-group separation, Figure 3 reports standardized mean differences at pretest and posttest. Unlike within-group standardized-change effects, these Cohen's d values can be calculated from the reported group means and standard deviations without requiring the unreported pre-post covariance.



Note. Approximate 95% CIs are based on reported group means, SDs, and $n = 30$ per group.

Figure 3. Between-group standardized mean differences (Cohen's d) at pretest and posttest with approximate 95% confidence intervals.

Figure 3 shows small standardized baseline differences for both outcomes, with confidence intervals crossing zero, followed by very large posttest separation favoring Smart Coaching. This estimation-focused display complements the interaction tests while avoiding unsupported repeated-measures standardization. Figure 4 summarizes relative percentage improvement to facilitate cross-scale comparison between CAPL-2 and FMS. Because the two outcomes use different score ranges, relative change provides an intuitive descriptive supplement rather than a substitute for the inferential analyses.

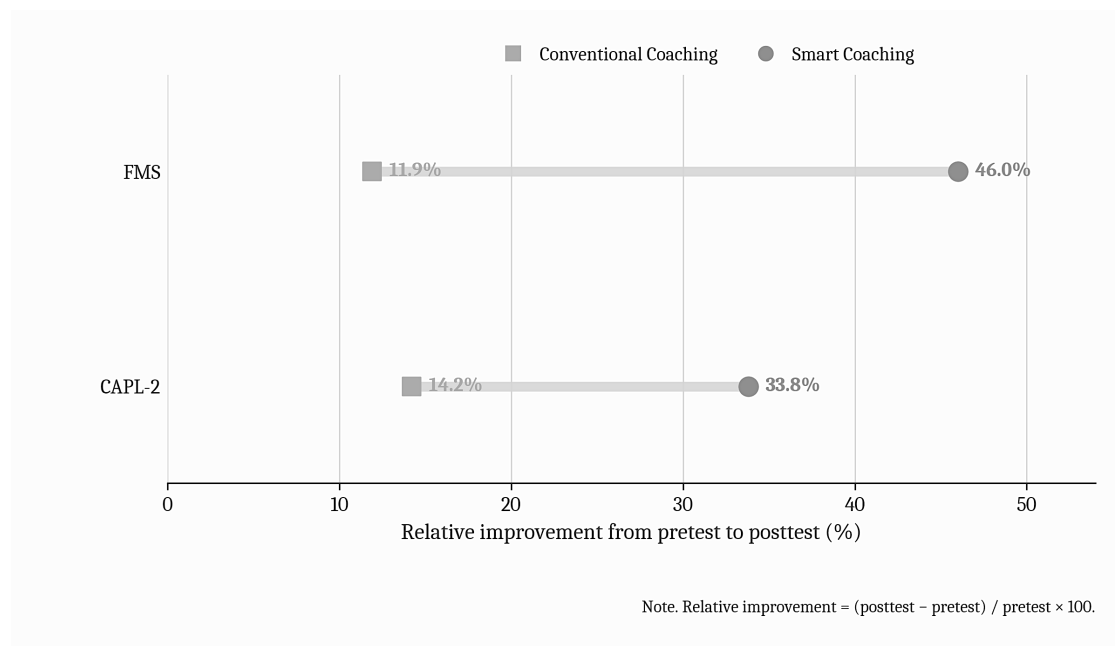


Figure 4. Relative pretest-to-posttest improvement by outcome and coaching condition.

The Smart Coaching condition showed greater relative improvement for both outcomes, with the contrast especially pronounced for FMS. Interpret these percentages descriptively, as percentage change can be sensitive to baseline scale position.

Discussion

This study examined whether an eight-week Smart Coaching condition supported by markerless motion analysis was associated with greater change in physical literacy and movement quality than conventional coaching. The interaction tests supported both hypotheses: the Smart Coaching group showed larger gains in the CAPL-2-based composite and FMS total score. The interaction effect sizes were large (partial $\eta^2 = .454$ and $.571$), but their interpretation must remain anchored to the nonrandomized, single-site design. The results demonstrate differential short-term change under the observed conditions; they do not establish that motion analysis alone caused the improvements.

The physical-literacy result is notable because the outcome is broader than technical movement competence. Meta-analytic evidence indicates that physical-literacy interventions can improve aggregate physical literacy and its constituent domains, with physical competence often showing the largest effects and motivation/confidence showing smaller but meaningful changes (Carl et al., 2022). More recent school-based evidence similarly emphasizes that effective physical-literacy interventions are multidimensional rather than purely motoric (Jerebine et al., 2024). The present composite gain is directionally consistent with this literature, but domain-level scores were unavailable; therefore, we cannot conclude that the markerless-feedback component independently improved motivation, confidence, knowledge, or daily activity behavior.

A plausible mechanism for part of the physical-literacy change is improved access to augmented feedback. Visual and technology-supported feedback can make features more salient, help learners compare intended and performed actions, and provide a basis for corrective practice. Reviews in physical education and sport report beneficial effects of visual or augmented feedback, although effectiveness depends on the task, delivery schedule, and learner (Möding et al., 2022; Tissera et al., 2022; van der Veer et al., 2022). In resistance-training contexts, feedback has also been associated with improved kinematic performance and motivation (Weakley et al., 2023). These findings make the Smart Coaching pathway theoretically credible, but they do not prove that the same mediators operated in the current participants.

The FMS result is more directly aligned with the intervention's movement-correction logic. The Smart Coaching group gained 5.23 points compared with 1.43 points in Conventional Coaching. Systematic review evidence indicates that structured exercise and corrective programs can increase

FMS total scores, including in athletic or otherwise active populations, but intervention contents vary substantially across studies (Clark et al., 2022; Huang et al., 2022). The magnitude observed here is therefore best understood as a large study-specific short-term change rather than a universal expected effect of motion-analysis coaching. Because FMS is a screening-style movement measure, the result should not be extrapolated to reduced injury risk without direct injury surveillance.

Markerless motion analysis may have increased the specificity of corrective information available to coaches, but current validation literature argues against treating all markerless outputs as laboratory-equivalent. Studies of OpenCap, deep-learning systems, and other markerless approaches demonstrate useful performance in selected tasks while also documenting joint-, plane-, and task-dependent error (Turner et al., 2024; Bae et al., 2024; Huang et al., 2024; Lima et al., 2024). Systematic reviews similarly show heterogeneous validity across systems and movement contexts (Yoma et al., 2025; Ogura et al., 2026). Accordingly, the present intervention should be conceptualized as data-supported coaching: markerless cues augment professional interpretation rather than replace it.

The adolescent and multisport context also matters. Motor competence in adolescence is associated with physical activity, fitness, and psychosocial characteristics, but these relationships are modest and influenced by age, sex, maturation, and measurement approach (Burton et al., 2023; Coe et al., 2024). The present sample spanned athletics, badminton, taekwondo, and soccer, which broadens ecological relevance but also introduces heterogeneity in training demands. Because the authors did not report sport distribution, sex, or maturation status by group, the analysis cannot rule out residual confounding. Future multisport trials should stratify or adjust for these variables and report participant flow and attendance in detail.

Measurement of physical literacy requires particular caution. CAPL-2 was originally developed for ages 8-12, whereas this sample extended to age 15 (Longmuir et al., 2018). Assessment reviews and adolescent-specific tools show that validity does not automatically transfer across age groups and contexts (Barnett et al., 2023; Mota et al., 2022, 2023). The CAPL-2-derived score is therefore interpreted as a continuous adapted study composite rather than an age-normative classification. Replication should use an instrument validated specifically for the target adolescent age range or conduct formal cross-cultural and age validation before testing intervention effects.

Implications

From a coaching-practice perspective, the study supports further evaluation of field-deployable motion feedback within routine training. The relevant practical feature is not presumed affordability or autonomous artificial intelligence, neither of which was measured, but the capacity to transform recorded movement into interpretable cues that can inform corrective practice. Recent markerless systems can reduce physical infrastructure relative to laboratory motion capture, yet implementation still depends on camera setup, computational workflow, coach competence, data governance, and system-specific measurement quality (Aderinola et al., 2023; Adlou et al., 2025; Noorbhai et al., 2025). Coaches and organizations should therefore validate the chosen system for the intended movement tasks before using its outputs for high-stakes decisions.

Research Contribution

The study contributes by integrating markerless motion-analysis cues into a coaching-feedback pathway and evaluating two complementary developmental outcomes in novice athletes: a multidimensional physical-literacy composite and functional movement quality. The contribution is not a claim that markerless technology independently causes athlete development, but evidence that a data-supported coaching condition can be associated with greater short-term changes than conventional coaching under these field conditions.

Limitations

Several limitations constrain inference. First, allocation was nonrandomized, and the administrative allocation mechanism was not documented, leaving selection bias possible. Second, the sample was small, multisport, and drawn from one Indonesian regional training center. Third, group-specific sex, age means, sport distribution, maturation status, and attendance were unavailable. Fourth, the authors did not document capture frequency, coaching progression, system

architecture, camera geometry, calibration, intervention fidelity, or assessor blinding in sufficient detail for full technological replication. Fifth, the CAPL-2 age extension lacks study-specific validation, and domain-level CAPL-2 data were unavailable. Sixth, the authors did not analyze markerless biomechanical variables longitudinally, so they cannot test the proposed feedback mechanism directly. Seventh, the eight-week design provides no evidence about retention, injury outcomes, sustained activity behavior, or long-term athlete development. These limitations warrant cautious interpretation despite the large observed interaction effects.

Suggestions

Future research should use randomized or clearly matched allocation, preregistered intervention protocols, baseline-adjusted sensitivity analyses or mixed-effects models, and participant-level reporting with attrition and adherence. Studies should archive the full motion-analysis pipeline, including hardware, software/version, pose-estimation model, calibration, camera geometry, processing rules, and feedback frequency, and report system-specific reliability for the tasks used. Mechanistic evaluation should analyze domain-level physical-literacy outcomes alongside prespecified kinematic variables and assess whether changes in movement cues mediate changes in FMS or physical competence. Follow-up testing is also needed to determine whether short-term gains persist after withdrawing technology-supported feedback.

CONCLUSION

In this eight-week, single-site, nonrandomized quasi-experiment, novice athletes receiving Smart Coaching supported by markerless motion analysis showed larger pretest-to-posttest increases in a CAPL-2-based physical-literacy composite and FMS total score than athletes receiving conventional coaching. The two significant Time $\times$ Group interactions support the conclusion that short-term change patterns differed between training conditions. The study does not directly demonstrate faster motor learning, improved injury prevention, sustained physical-activity behavior, or long-term athlete development. Its main contribution is evidence that integrating objective movement cues into a structured coaching-feedback process is a promising technology-supported approach that merits more rigorous, fully reproducible testing.

ACKNOWLEDGMENT

Acknowledgment details are withheld to preserve double-blind peer review. The study received institutional research funding; funder identity is provided separately in the journal submission metadata.

AUTHOR CONTRIBUTION STATEMENT

For blinded peer review, author names are represented by position in the author list. RD conceptualization, methodology, project administration, investigation, data curation, formal analysis, writing-original draft, writing-review and editing, and correspondence; AN conceptualization, methodology, supervision, validation, formal analysis, and writing-review and editing; EF software checking, motion-analysis system development, data curation, visualization, and formal analysis. BSF investigation, resources, data collection, validation, and writing-review and editing; MRD investigation, resources, validation, visualization, and writing-review and editing. All authors reviewed and approved the manuscript and accept accountability for the integrity of the work.

AI DISCLOSURE STATEMENT

The authors used OpenAI ChatGPT to support English-language editing, editorial organization, and consistency checking. All numerical results, references, and interpretations are the responsibility of the authors, who retain full responsibility for the manuscript.

CONFLICTS OF INTEREST

Conflict-of-interest disclosure is withheld from the blinded manuscript to preserve author anonymity and is provided through the journal's submission process.

ETHICAL APPROVAL

The authors' institutional research ethics committee approved the protocol (approval no. 025/OR/PL.1/07/2025) before participant recruitment and data collection. Because participants were minors, each athlete provided written agreement to participate, and a parent or legal guardian provided written informed consent. No participant-identifying images or video frames are reported.

REFERENCES

- Aderinola, T. B., Younesian, H., Whelan, D., Caulfield, B., & Ifrim, G. (2023). Quantifying jump height using markerless motion capture with a single smartphone. *IEEE Open Journal of Engineering in Medicine and Biology*, 4, 109-115. <https://doi.org/10.1109/OJEMB.2023.3280127>
- Adlou, B., Wilburn, C., & Weimar, W. (2025). Motion capture technologies for athletic performance enhancement and injury risk assessment: A review for multi-sport organizations. *Sensors*, 25(14), 4384. <https://doi.org/10.3390/s25144384>
- Bae, K., Lee, S., Bak, S.-Y., Kim, H. S., Ha, Y., & You, J. H. (2024). Concurrent validity and test reliability of the deep learning markerless motion capture system during the overhead squat. *Scientific Reports*, 14, 29462. <https://doi.org/10.1038/s41598-024-79707-2>
- Barnett, L. M., Jerebine, A., Keegan, R., Watson-Mackie, K., Arundell, L., Ridgers, N. D., Salmon, J., & Dudley, D. (2023). Validity, reliability, and feasibility of physical literacy assessments designed for school children: A systematic review. *Sports Medicine*, 53(10), 1905-1929. <https://doi.org/10.1007/s40279-023-01867-4>
- Burton, A. M., Cowburn, I., Thompson, F., Eisenmann, J. C., Nicholson, B., & Till, K. (2023). Associations between motor competence and physical activity, physical fitness and psychosocial characteristics in adolescents: A systematic review and meta-analysis. *Sports Medicine*, 53(11), 2191-2256. <https://doi.org/10.1007/s40279-023-01886-1>
- Carl, J., Barratt, J., Wanner, P., Töpfer, C., Cairney, J., & Pfeifer, K. (2022). The effectiveness of physical literacy interventions: A systematic review with meta-analysis. *Sports Medicine*, 52(12), 2965-2999. <https://doi.org/10.1007/s40279-022-01738-4>
- Clark, S. C., Rowe, N. D., Adnan, M., Brown, S. M., & Mulcahey, M. K. (2022). Effective interventions for improving Functional Movement Screen scores among 'high-risk' athletes: A systematic review. *International Journal of Sports Physical Therapy*, 17(2), 131-138. <https://doi.org/10.26603/001c.31001>
- Coe, D. P., Post, E. M., Fitzhugh, E. C., Fairbrother, J. T., & Webster, E. K. (2024). Associations among motor competence, physical activity, perceived motor competence, and aerobic fitness in 10-15-year-old youth. *Children*, 11(2), 260. <https://doi.org/10.3390/children11020260>
- Cronin, N. J., Walker, J., Tucker, C. B., Nicholson, G., Cooke, M., Merlino, S., & Bissas, A. (2024). Feasibility of OpenPose markerless motion analysis in a real athletics competition. *Frontiers in Sports and Active Living*, 5, 1298003. <https://doi.org/10.3389/fspor.2023.1298003>
- Fitton Davies, K., Sacko, R. S., Lyons, M. A., & Duncan, M. J. (2022). Association between Functional Movement Screen scores and athletic performance in adolescents: A systematic review. *Sports*, 10(3), 28. <https://doi.org/10.3390/sports10030028>
- Huang, J., Zhong, M., & Wang, J. (2022). Effects of exercise-based interventions on functional movement capability in untrained populations: A systematic review and meta-analysis. *International Journal of Environmental Research and Public Health*, 19(15), 9353. <https://doi.org/10.3390/ijerph19159353>
- Huang, T., Ruan, M., Huang, S., Fan, L., & Wu, X. (2024). Comparison of kinematics and joint moments calculations for lower limbs during gait using markerless and marker-based motion capture. *Frontiers in Bioengineering and Biotechnology*, 12, 1280363. <https://doi.org/10.3389/fbioe.2024.1280363>
- Jerebine, A., Arundell, L., Watson-Mackie, K., Keegan, R., Jurić, P., Dudley, D., Ridgers, N. D., Salmon, J., & Barnett, L. M. (2024). Effects of holistically conceptualised school-based interventions on children's physical literacy, physical activity, and other outcomes: A systematic review. *Sports Medicine - Open*, 10(1), 105. <https://doi.org/10.1186/s40798-024-00766-w>

- Lima, Y. L., Collings, T., Hall, M., Bourne, M. N., & Diamond, L. E. (2024). Validity and reliability of trunk and lower-limb kinematics during squatting, hopping, jumping and side-stepping using OpenCap markerless motion capture application. *Journal of Sports Sciences*, 42(19), 1847-1858. <https://doi.org/10.1080/02640414.2024.2415233>
- Longmuir, P. E., Gunnell, K. E., Barnes, J. D., Belanger, K., Leduc, G., Woodruff, S. J., & Tremblay, M. S. (2018). Canadian Assessment of Physical Literacy Second Edition: A streamlined assessment of the capacity for physical activity among children 8 to 12 years of age. *BMC Public Health*, 18, 1047. <https://doi.org/10.1186/s12889-018-5902-y>
- Moon, J., Webster, C. A., Stodden, D. F., Brian, A., Mulvey, K. L., Beets, M., Egan, C. A., McIntosh, L. I. F., Merica, C. B., & Russ, L. (2024). Systematic review and meta-analysis of physical activity interventions to increase elementary children's motor competence: A comprehensive school physical activity program perspective. *BMC Public Health*, 24, 826. <https://doi.org/10.1186/s12889-024-18145-1>
- Mota, J., Martins, J., & Onofre, M. (2022). Portuguese Physical Literacy Assessment - Observation (PPLA-O) for adolescents (15-18 years) from grades 10-12: Development and initial validation through item response theory. *Frontiers in Sports and Active Living*, 4, 1033648. <https://doi.org/10.3389/fspor.2022.1033648>
- Mota, J., Martins, J., & Onofre, M. (2023). Portuguese Physical Literacy Assessment Questionnaire (PPLA-Q) for adolescents: Validity and reliability of the psychological and social modules using Mokken scale analysis. *Perceptual and Motor Skills*, 130(3), 958-983. <https://doi.org/10.1177/00315125231159688>
- Möding, M., Woll, A., & Wagner, I. (2022). Video-based visual feedback to enhance motor learning in physical education-a systematic review. *German Journal of Exercise and Sport Research*, 52, 447-460. <https://doi.org/10.1007/s12662-021-00782-y>
- Noorbhai, H., Moon, S., & Fukushima, T. (2025). A conceptual framework and review of multi-method approaches for 3D markerless motion capture in sports and exercise. *Journal of Sports Sciences*, 43(12), 1167-1174. <https://doi.org/10.1080/02640414.2025.2489868>
- O'Brien, W., Khodaverdi, Z., Bolger, L., Tarantino, G., Philpott, C., & Neville, R. D. (2022). The assessment of functional movement in children and adolescents: A systematic review and meta-analysis. *Sports Medicine*, 52(1), 37-53. <https://doi.org/10.1007/s40279-021-01529-3>
- Ogura, A., Florio, E., Wileman, T. M., Dennison, L., & Psarakis, M. (2026). Are we there yet? A systematic review and meta-analysis of the validity and reliability of automated markerless motion capture systems during jumping tasks. *Journal of Sports Sciences*, 44(10), 1275-1295. <https://doi.org/10.1080/02640414.2025.2589689>
- Philipp, N. M., Fry, A. C., Mosier, E. M., Cabarkapa, D., Nicoll, J. X., & Sontag, S. A. (2024). Biological reliability of a movement analysis assessment using a markerless motion capture system. *Frontiers in Sports and Active Living*, 6, 1417965. <https://doi.org/10.3389/fspor.2024.1417965>
- Söker, N., Schamerowski, J., Zech, A., Steib, S., & Rahlf, L. (2025). Long-term athlete development in schools: A systematic review and narrative synthesis. *Frontiers in Sports and Active Living*, 7, 1706060. <https://doi.org/10.3389/fspor.2025.1706060>
- Tissera, K., Orth, D., Huynh, M., & Benson, A. C. (2022). The impact of augmented feedback (and technology) on learning and teaching cricket skill: A systematic review with meta-analysis. *PLOS ONE*, 17(12), e0279121. <https://doi.org/10.1371/journal.pone.0279121>
- Turner, J. A., Chaaban, C. R., & Padua, D. A. (2024). Validation of OpenCap: A low-cost markerless motion capture system for lower-extremity kinematics during return-to-sport tasks. *Journal of Biomechanics*, 171, 112200. <https://doi.org/10.1016/j.jbiomech.2024.112200>
- van der Veer, I. P. A., Verbecque, E., Rameckers, E. A. A., Bastiaenen, C. H. G., & Klingels, K. (2022). How can instructions and feedback with external focus be shaped to enhance motor learning in children? A systematic review. *PLOS ONE*, 17(8), e0264873. <https://doi.org/10.1371/journal.pone.0264873>
- Varcin, F., & Boockock, M. G. (2026). The accuracy, validity and reliability of Theia3D markerless motion capture for studying the biomechanics of human movement: A systematic review. *Artificial Intelligence in Medicine*, 173, 103332. <https://doi.org/10.1016/j.artmed.2025.103332>

- Weakley, J., Cowley, N., Schoenfeld, B. J., Read, D. B., Timmins, R. G., García-Ramos, A., & McGuckian, T. B. (2023). The effect of feedback on resistance training performance and adaptations: A systematic review and meta-analysis. *Sports Medicine*, 53(9), 1789-1803. <https://doi.org/10.1007/s40279-023-01877-2>
- Welsby, E., Hordacre, B., Hobbs, D., Bouckley, J., Ward, E., & Hillier, S. (2024). Evaluating the influence of feedback on motor skill learning and motor performance for children with developmental coordination disorder: A systematic review. *Frontiers in Pediatrics*, 12, 1327445. <https://doi.org/10.3389/fped.2024.1327445>
- Yeadon, M. R., & Pain, M. T. G. (2023). Fifty years of performance-related sports biomechanics research. *Journal of Biomechanics*, 155, 111666. <https://doi.org/10.1016/j.jbiomech.2023.111666>
- Yoma, M., Llurda-Almuzara, L., Herrington, L., & Jones, R. (2025). Reliability and validity of lower extremity and trunk kinematics measured with markerless motion capture during sports-related and functional tasks: A systematic review. *Journal of Sports Sciences*, 43(17), 1703-1730. <https://doi.org/10.1080/02640414.2025.2518359>